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# The stochastic location-routing problem with parallel truck–drone operations for humanitarian aid delivery

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ABSTRACT

Timely response in the aftermath of a disaster is crucial to alleviate loss of life and suffering. Timeliness of relief may be hampered by road network disruptions caused by the disaster, such as damage to road segments or debris covering the roads. The use of drones simultaneously with trucks can potentially help overcome issues around network disruptions and achieve more timely delivery of post-disaster aid. In an effort to shed more light into this potential, we address the problem of network design for parallel truck–drone operations by depot location prior to the disaster and routing of the vehicles in its aftermath. We incorporate the uncertainty on network disruption by modelling this problem as a two-stage stochastic program, which proves computationally challenging to solve to optimality for real-life disaster scenarios. Consequently, we propose a tailored heuristic based on variable neighbourhood search to find high-quality solutions efficiently. Our computational results on randomly generated instances and a case study from the 2011 Van Earthquake in Turkiye demonstrate the effectiveness of the heuristic, the benefits of employing both trucks and drones, and the significance of accounting for uncertainties in pre-disaster planning.

## 1. Introduction

In the aftermath of large-scale disasters (e.g., earthquakes, hurricanes, terrorist attacks), timely response to the needs of the beneficiaries is crucial. Logistics accounts for 80 percent of disaster relief efforts (Kovács & Spens, 2007), with infrastructure networks including roads, power, and telecommunications playing a vital role in disaster response by facilitating search-and-rescue operations, evacuation, and distribution of relief items. However, these networks are also vulnerable to the effects of disasters. Road networks, in particular, can be compromised by collapsed segments and bridges, blockage due to debris, and damage to critical facilities. As ground vehicles are the primary means of transportation for humanitarian aid, such disruptions can isolate beneficiary locations from relief aid vehicles (Celik, 2016). Originally designed for military purposes, unmanned air vehicles (UAVs) offer significant potential to overcome these challenges by being able to reach isolated areas in a timely manner.

UAVs (or drones) have already found a wide range of applications outside humanitarian aid such as terrain mapping, agricultural spraying, wildlife monitoring and conservation (Otto et al., 2018). In the field of logistics, many companies have started carrying out part of their last-mile delivery operations using drones (Amazon, 2023; Walmart, 2024). Despite the initially restrictive regulations on UAV use due to privacy, security and operational concerns, enhancements in technology and familiarity with their potential have meant that these restrictions are being gradually relaxed over the recent years (Young, 2024). In the United States, the Federal Aviation Administration has authorised companies including UPS, uAvionics, and Zipline to operate beyond visual line of sight and use the collected data in their ongoing policy making activities for drones (FAA, 2023a, 2023b). In the United Kingdom, the Civil Aviation Authority has recently put forward similar measures to expand drone use for commercial and non-commercial activities. (The Guardian, 2024).

Drones are also being increasingly used in humanitarian relief. For example, as one of the earliest adopters in the humanitarian sector, The United Nations World Food Programme uses drones for food aid delivery in areas with significant access limitations (World Food Programme, 2023). The Uganda Ministry of Health, in collaboration with other non-governmental organisations, uses drones to send HIV/AIDS medicines to island communities with high HIV prevalence around Lake Victoria (The Guardian, 2021). The drone company Matternet transports biomedical supplies between hospitals in Zurich, and lab samples in Berlin, and has launched operations for medical supply delivery between facilities in Abu Dhabi (Matternet, 2024).

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Despite their advantages in improving accessibility in disrupted networks, drones are also limited in their range and battery life. Consequently, most humanitarian aid delivery involving drones is performed in conjunction with ground vehicles, particularly trucks. A growing body of literature on routing trucks and drones for last-mile deliveries in both for-profit and humanitarian applications has considered various ways in which such operations can be performed, such as independent deliveries by both vehicles from depots, synchronised operations where the drone travels on the truck and both perform deliveries, and one vehicle supporting the delivery operations carried out by the other (Otto et al., 2018). On the other hand, there is limited research on optimising parallel delivery of humanitarian aid by trucks and drones, despite their widespread use in practice.

The efficiency and effectiveness of truck–drone delivery operations rely heavily on the supporting transportation network. Depots play a crucial role in this support by storing relief items, acting as a base for launching and recharging drones, loading and unloading trucks, and coordinating the deliveries by trucks and drones. In designing the network, the decision of where to locate these depots is an important one due to its direct influence the timeliness of aid deliveries. Pre-disaster location decisions save time for subsequent relief delivery by enabling immediate dispatch of vehicles, compared to making such decisions during the response stage. However, these pre-disaster decisions must also account for disaster-related uncertainties, such as the impact on the transportation network, as the type, location, and magnitude of the disaster are not known in advance.

In this paper, we address the decisions of pre-disaster depot location and post-disaster relief aid delivery by trucks and drones that operate independently, where the post-disaster travel times on the road network are uncertain. Contrary to the literature on truck–drone routing where the main objective is to minimise makespan or operational costs, we prioritise the timeliness of deliveries by considering an objective that aims to minimise the expected average arrival time to all beneficiaries. The objective reflects the cumulative urgency of aid delivery across multiple communities, especially in the immediate aftermath of a disaster where early access to relief can prevent deterioration of health and safety. To the best of our knowledge, this is the first study that considers such a latency-based objective in a stochastic location-routing problem involving the parallel use of trucks and drones for last-mile humanitarian aid delivery. Our problem definition and the network structure lead to a novel two-stage mixed integer stochastic programming model that can only solve small-sized instances to optimality using commercial solvers. To overcome this challenge, we propose a variable neighbourhood search heuristic, which produces high-quality solutions for larger instances within reasonable time limits. We use a case study from the 2011 Van Earthquake in Turkiye to understand the structure of resulting tours, as well as to compare our approach to benchmarks involving only trucks, no assessment on road network damage, and separate location and routing decisions. The computational results underline the potential effectiveness of our approach and present insights that can be of benefit to decision makers.

The remainder of the paper is organised as follows: Section 2 presents a literature review on truck–drone delivery models and stochastic location-routing problems in humanitarian operations. Section 3 outlines the problem definition and provides the two-stage stochastic mixed integer programming model. In Section 4, we introduce a problem-tailored variable neighbourhood search heuristic to solve the instances of practical size. Section 5 details the computational experiments and provides the managerial insights from our findings. Concluding remarks and further research directions are presented in Section 6.

## 2. Literature review

This paper is positioned at the intersection of truck–drone delivery problems, location-routing problems with heterogeneous fleets and multiple vehicle types, and stochastic location-routing problems in humanitarian operations. In this section, we present an overview of these three streams and the contributions of our work to the literature.

### 2.1. Truck–drone delivery problems

Due to the increase in application areas and technological advances for using drones in last-mile delivery, there is increased interest on truck–drone operations in the extant literature. An extensive review to date on the combined operations of drones with other vehicles in the non-military context is provided in Otto et al. (2018). Other reviews that discuss truck–drone combined operations include Chung et al. (2020) on optimisation methods, Macrina et al. (2020) on parcel delivery, Rojas Viloria et al. (2021) on relevant vehicle routing problems, Moshref-Javadi and Winkenbach (2021) on general logistics applications, and Yucesoy et al. (2024) on their use in disaster response. In their review of facility location for drone delivery, Dukkanci et al. (2023) point out that 95% of the literature on facility location for truck–drone deliveries considers commercial applications, with only one paper (Chowdhury et al., 2017) in humanitarian response. Our paper aims to bridge this gap in the literature.

The literature on truck–drone delivery can be divided into three categories in terms of the delivery structure.

Models in the first category consider a primary vehicle type (truck or drone) that makes the deliveries, while the other vehicle type supports the operations. In Ferrandez et al. (2016), drones make the deliveries, while trucks serve as launching stations, so that the effective coverage area of drones is expanded. Scott and Scott (2019) propose a structure in which drones make the last-mile deliveries, whereas trucks supply the humanitarian aid to drone stations from a warehouse. They formulate a location-routing problem to determine the locations of the warehouse and drone nests. Zhou et al. (2023) study a similar structure, locating drone stations in 2-D space. They model the problem as a constrained nonlinear optimisation, and present a solution approach based on particle swarm optimisation with a subordinate dynamic programme, using healthcare delivery as a case study. Variations of such structure are previously studied by Kim and Moon (2018) and Schermer et al. (2019) to extend the drone service range. McCunney and Cauwenberghe (2019) and Dayarian et al. (2020) propose drone-supported truck deliveries, with the drones resupplying the trucks directly or supplying the depots within a same-day delivery framework.

The second group of delivery structures involves a synchronised truck–drone delivery model where the drone travels with the truck and both vehicles perform deliveries. The routing problem corresponding to this version of truck–drone tandem delivery is more commonly known as the Flying Sidekick Travelling Salesman Problem (Murray & Chu, 2015) or the TSP with Drones (Agatz et al., 2018). In this case, the drone travels with a truck to extend its service area, launches from the truck, delivers the items, and lands back on the truck again. Meanwhile, after the drone takes off, the truck continues its tour and visits the next demand location, and the two vehicles meet at one of the demand locations. Moshref-Javadi and Lee (2017) consider tandem deliveries for situations such as disaster relief and emergencies, using a latency-based objective. Although their setting is humanitarian, their model does not incorporate any uncertainties and focuses solely on the routing problem. Yin et al. (2023) also address the routing problem in humanitarian operations under demand and travel time uncertainties. They develop a robust optimisation model using a budgeted uncertainty set and propose a branch-and-price-and-cut algorithm, applying it to a real earthquake case study. Mirzapour Al-e Hashem et al. (2024) apply tandem delivery to a postal delivery system, considering weather-related uncertainties. They model the problem as a robust optimisation and develop a hybrid metaheuristic solution approach. Kang and Lee (2021) utilise heterogeneous drones with different limitations and develop an exact algorithm for the routing problem using logic-based Bender decomposition. Morim et al. (2024) extend truck–drone tandem routing by incorporating robot deliveries. When a truck visits a robot station, it activates the station, and pre-positioned robots then perform return deliveries. Gómez-Lagos et al. (2021) introduce a variant of

truck–drone tandem delivery, where the truck can park at a non-customer location during its tour, allowing drones to make return trips from either demand nodes or parking lots. They apply a GRASP metaheuristic to minimise makespan, incorporating truck routing, drone trips, and parking lot selection in the procedure. Similarly, Montaña et al. (2022) allow drone launches from truck but only when the truck is at a parking lot, and also include parking lot selection in their model. There is substantial literature on synchronised models; the interested reader is referred to Moshref-Javadi et al. (2021) for a comprehensive literature review to date, and to Dukkanci et al. (2024) for facility location considerations.

Our study belongs to the third category of truck–drone delivery operations, where trucks and drones perform deliveries in parallel from the depots. Murray and Chu (2015) first formalise this problem as the Parallel Drone Scheduling Travelling Salesman Problem (PDSTSP). They formulate a mixed integer program and develop a two-phase heuristic to solve it for large-sized instances. For a single pair of truck and drone, Mbiadou Saleu et al. (2018) develop an iterative two-step heuristic and Dell'Amico et al. (2020) propose matheuristic methods. Nguyen et al. (2022) extend the PDSTSP to involve multiple trucks and drones and propose a heuristic algorithm. Kloster et al. (2023) consider a variant of parallel operations where trucks make deliveries along their tours, and when a truck visits a pre-positioned robot station, the station is activated to enable return-trip deliveries by drones or robots. Some of the earlier studies categorised under the first category of truck–drone operations, those using drone nests or stations (e.g., Kim & Moon, 2018; Schermer et al., 2019; Scott & Scott, 2019; Zhou et al., 2023), can also be interpreted as variants of the PDSTSP. In these models, trucks support drone operations by transporting packages to drone stations, from which drones conduct last-mile deliveries to extend their range. While these studies involve location-routing decisions due to the inclusion of station location choices, they do not incorporate uncertainty in their models.

Only a few studies on the PDSTSP variants address uncertainties. Rezaei Kallaj et al. (2023) tackle blood delivery by trucks and drones in disaster scenarios, accounting for supply and demand uncertainties using a two-stage stochastic model and particle swarm optimisation for its solution. Van Steenbergen et al. (2023) address travel time uncertainties in post-disaster relief with a stochastic dynamic program and deep reinforcement learning. Studies incorporating network design into PDSTSP are also scarce. Cokyasar et al. (2021) address the decision of locating automated battery swapping machines in a commercial delivery setting and solve the problem with a derivative-supported cutting-plane method. Chowdhury et al. (2017) uniquely integrate network design and uncertainty, using drones when trucks are unavailable and employing a probabilistic approach to road access, with routes determined through a continuous approximation method.

### 2.2. Location-routing problems with heterogeneous fleets and multiple vehicle types

A significant amount of literature exists on the Location-Routing Problem (LRP). Comprehensive reviews on the problem are given in the survey papers by Nagy and Salhi (2007), Prodhon and Prins (2014), Drexl and Schneider (2015), Schneider and Drexl (2017), Albareda-Sambola and Rodríguez-Pereira (2019), and Mara et al. (2021). The work in this paper is a special case of the LRP with heterogeneous fleets and multiple vehicle types. For the sake of brevity, we focus our attention to the papers in this stream of literature.

Due to its practical applicability, the LRP with heterogeneous vehicles has gained significant attention in the literature. Among the recent exact approaches to tackle the problem, Çalık et al. (2021) use a Benders decomposition approach to solve the resulting mixed integer program, whereas (Wang et al., 2025) devise a branch-and-price-and-cut algorithm using bidirectional labelling for a variant of the problem that minimises emission and operational costs. Due to the inherent complexity of solving the problem to optimality, heuristic and metaheuristic approaches are more commonly used. Among these, Koç et al. (2016) use an evolutionary algorithm for the problem involving time windows, Zhao et al. (2018) propose a cooperative approximation approach, Amiri et al. (2019) use Lagrangean decomposition for a multi-echelon application in offshore oil and gas supply chains, and Wang et al. (2019) employ multiple neighbourhood search algorithms for a multiobjective version of the problem.

The LRP with multiple vehicle types is relevant to the problem addressed in this paper due to the involvement of multiple networks for vehicle routing. Klibi et al. (2010) consider a multiperiod version of the Stochastic LRP where customers can be served by full or less-than-truckload shipments, using a tabu search heuristic to find efficient solutions. Snoeck et al. (2018) make network design and fleet planning decisions for Coca Cola in Bogota using a linear approximation of the resulting mathematical model. Zhou et al. (2019) use a hybrid genetic algorithm and simulated annealing approach on a variant with simultaneous home delivery and customer pickup. De Maio et al. (2023) focus on an application in the dairy industry with stochastic demands and make use of an indifference zoning approach to solve the problem. Ozyavas et al. (2025) consider a parcel locker delivery system with trucks at the upstream and bike deliveries at the downstream of the chain, solving the problem using a branch-and-price algorithm.

### 2.3. Stochastic location-routing problems in humanitarian operations

In this section, we provide an overview of relevant applications of the Stochastic Location-Routing Problem in humanitarian operations. Interested reader can also refer to Temiz et al. (2025) for a recent review on stochastic location-routing problems, addressing both humanitarian and other application areas.

Most studies on the Stochastic Location-Routing Problem in humanitarian operations involves selection of pre-defined routes in the second-stage, rather than explicit routing. Ukkusuri and Yushimito (2008) address a location-routing problem with inventory pre-positioning under route failures, selecting reliable routes based on failure probabilities, and using the set of reliable routes to make location decisions. Mete and Zabinsky (2010) study depot location with pre-positioning to serve hospitals in case of a disaster, using a two-stage stochastic program to handle demand and travel time uncertainties. Bozorgi-Amiri and Khorsi (2016) define a multi-period multi-objective problem with demand, travel time and cost uncertainties. Aslan and Çelik (2019) consider another version of the Stochastic Location-Routing Problem with pre-positioning under a latency-based objective in the second-stage, solving the problem to optimality using valid inequalities. Chang et al. (2017) formulate a robust optimisation model for a variant of the problem with demand and travel speed uncertainty. Although this study explicitly considers construction of tours, it does so only for the worst-case scenario. In a different vein, Ahmadi et al. (2015) incorporate route construction into the first-stage decisions considering the road failures and inflated travel times (via decreased travel speed), while the second-stage calculates constraint violations for relief times for each scenario.

Although the use of pre-defined routes increases the tractability of the resulting Stochastic Location-Routing Problem, it also decreases the effectiveness of relief transportation by limiting the potential set of routes that can be used. Only a limited number of studies in the literature for the Stochastic Location-Routing Problem in humanitarian operations explicitly construct routes after uncertainty realisations. Among these, Caunhye et al. (2016) address pre-positioning and routing of emergency supplies under demand and travel time uncertainties, and use robust optimisation by minimising the cost of setting up depots and the worst-case results among damage scenarios. Sabouhi et al. (2021) also construct the routes in the second-stage considering demand uncertainty and road failure. Zhong et al. (2020) consider demand uncertainty and decision maker's risk attitude. They implement a genetic algorithm for the resulting bi-objective problem, using cost and the summation of arrival times at demand nodes as objectives.

### 2.4. Contributions of this study

The work in this paper makes important contributions to the methodology by means of a novel modelling and solution approach. It also contributes to the practice through the managerial insights it drives through the computational experiments.

From a methodological perspective, this study introduces a novel mathematical formulation for a stochastic location-routing problem with multiple vehicle types, latency-based objectives, and a sparse network lacking triangle inequality. These problem characteristics deem impossible the use of classical decision variable definitions and subtour elimination constraints typically used in the location-routing literature. To address this, we propose a flow-based formulation that accommodates multiple visits to nodes and develop a new set of subtour elimination constraints tailored for this structure. Moreover, we design a variable neighbourhood search (VNS) metaheuristic that jointly makes depot location and routing decisions across scenarios. Unlike prior work, our approach determines scenario-specific routes directly, without relying on adjustments of a single scenario-independent route. To the best of our knowledge, this is the first such application of VNS in this context.

On the practical side, the study offers insights into the design of post-disaster relief logistics using mixed fleets of trucks and drones under uncertainty. Our results demonstrate the value of integrating these vehicle types, highlight the importance of scenario-dependent planning over static or rule-based strategies, and quantify the operational impact of needs assessment. The findings also inform policy and design decisions by identifying feasible ranges for drone endurance and speed, thereby supporting the effective deployment of hybrid delivery systems in humanitarian settings.

## 3. Problem definition and mathematical model

We start this section by describing the problem environment, followed by our assumptions that result in the definition of a stochastic location-routing problem with parallel truck and drone deliveries and travel time uncertainties. In the last part of this section, we provide a two-stage stochastic mixed integer programming formulation for this problem.

### 3.1. Problem description

The problem at hand considers the decision of designing a relief distribution network in preparation for a large-scale disaster. Prior to the disaster, local authorities (e.g., municipalities) or federal/national agencies make the decisions on where to locate humanitarian depots equipped with parking stations, charging areas, and landing pads, to serve as hubs for the emergency vehicles (trucks and drones). These depots also serve as storage areas for relief items, which are generally pre-positioned before the disaster. Rather than building a new facility, opening a depot generally implies converting an existing public or private building, such as schools, town halls, sports halls, or warehouses (Aslan & Çelik, 2019). The decision of opening depots is generally subject to a fixed budget.

Our focus is on disasters where beneficiaries are located in smaller residential areas such as rural villages or districts. Demand of relief items is served in central areas of demand locations, such as town centres, and emergency assembly points such as schools and community centres. Relief items are single commodities or bundles, which may include medical supplies (first aid supplies, personal protective equipment), weather protection (emergency foil blankets, rain ponchos), or communication equipment (battery chargers, flashlights) (Noyan et al., 2016). In general, post-disaster demand amounts of such items are known a priori, as such quantities are proportional to the population of each demand location.

In the aftermath of a disaster, damaged infrastructure leads to blocked roads or longer traversal time on some of the road segments. As the structural impact of the potential disaster on the road network is uncertain, pre-disaster network design decisions need to be made with this uncertainty in mind. The potential effects of different disaster scenarios can be estimated using structural engineering models, such as HAZUS in the US (FEMA, 2022). Once the disaster hits, damage assessment on the ground transportation network is performed by a central decision maker to manage and coordinate timely search-and-rescue operations and humanitarian aid delivery. Geographical information systems, when integrated with satellite imagery, higher resolution drone images, and crowd-sourced data, can provide the road infrastructure condition in the aftermath.

Post-disaster routing of drones and trucks, as well as decisions on relief item distribution are carried out by a central authority, such as FEMA in the US and AFAD in Turkiye. These entities also coordinate the disaster response between local authorities, NGOs, and private parties. The aim of the central decision maker is to provide a timely delivery of relief items to all beneficiaries by means of an effective routing and scheduling of vehicles.

### 3.2. Modelling assumptions

For the sake of simplicity, we assume that the central decision-maker locates a pre-determined number of depots. The existence of a fixed budget (instead of a fixed number of depots) can be easily incorporated into our approach. We also assume that candidate depot locations coincide with those of demand locations. For simplicity, each open depot is assigned a single truck and a single drone. As in Murray and Chu (2015), trucks make a single tour from the depot, while drones can make multiple take-offs, with a single delivery each take-off. Truck and drone operations are in parallel; with trucks following a sequence of demand points in a route and drones conducting separate deliveries directly from the depot. We also assume a drone cannot visit more than one location per take-off due to operational constraints. Without loss of generality, we assume negligible waiting times for activities such as battery swapping and package loading. If needed, these can be incorporated into the travel times.

We consider a set of relief supplies whose demand at each location can be satisfied by a single delivery trip from either a truck or a drone. Examples of such commodities include the lightweight first-aid and sustenance packages of IFRC which weigh between 3–10 kilogrammes and can provide a sustenance package for a small to medium-sized villages (International Federation of Red Cross and Red Crescent Societies, 2018) and Interagency Emergency Health Kits of UNICEF, which provide medical kits for up to 100 children, and weighs less than 10 kilogrammes (UNICEF, 2025). These are small enough to be carried in a single trip to each demand location (e.g., Wingcopter GmbH, 2022; Zipline, 2023), and thus payload is not a concern.

Drones have many operational limitations, including performance in bad weather, payload capacity, battery duration, motion restrictions, etc. In line with previous studies (e.g., Agatz et al., 2018; Carlsson & Song, 2018; Murray & Chu, 2015), we incorporate only endurance as an explicit drone limitation in our model. Other operational limitations are excluded for three main reasons. First, payload is not a binding factor due to the size of relief commodities. Second, some factors such as weather-related performance or motion restrictions are harder to estimate in advance. Lastly, our model addresses a strategic-level problem, and the effects of such factors can be implicitly captured by calibrating the endurance parameter. We also assume constant drone performance throughout its flight such that the flight duration between a depot and a demand location is symmetric.

In the literature, road network disruptions are modelled in two alternative ways. Binary disruptions (e.g., Canbilen Sütiçen et al. 2023, Sabouhi et al. 2021, Salman and Yücel 2015) assume that once a road segment is disrupted, it is not traversable at all. Inflated travel times

**Table 1**
Notation used in the two-stage stochastic mixed integer programming model

| | |
|---|---|
| *Index Sets* | |
| $N$ | Nodes (except the sink node) |
| $s$ | Sink node |
| $N'$ | All nodes including the sink node |
| $E$ | Edges (excluding those incident to the sink node) |
| $E'$ | All edges including those incident to the sink node |
| $W$ | Candidate depot nodes |
| $D$ | Demand nodes |
| $W_i$ | Candidate depot nodes within the endurance limit for a return trip to the demand node $i \in D$ |
| $D_k$ | Demand nodes within the endurance limit for a return trip from depot $k \in W$, sorted in non-decreasing order of drone travel time to the depot |
| $i_m$ | Node index of the $m$th demand node in the ordered list $D_k$ |
| $\Omega$ | Disaster scenarios |
| *Deterministic Parameters* | |
| $v_{ik}$ | Round trip flight time of drone originating from depot $k \in W$ to demand node $i \in D$ |
| $n$ | Number of depots to be opened |
| *Stochastic Parameters* | |
| $t_{ij}(\xi)$ | Truck travel time on edge $(i,j) \in E$ under realisation $\xi \in \Omega$ |
| *First Stage Decision Variables* | |
| $x_k$ | $= \begin{cases} 1, & \text{if a depot is located at candidate depot node } k \in W \\ 0, & \text{otherwise} \end{cases}$ |
| *Second Stage Decision Variables* | |
| $y_{ijk}(\xi)$ | $= \begin{cases} 1, & \text{if a truck travels on edge } (i,j) \in E \text{ on a route originating from a depot } k \in W \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $c_{ik}(\xi)$ | $= \begin{cases} 1, & \text{if demand node } i \in D \text{ is assigned to a truck originating from a depot } k \in W \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $w_{ijk}(\xi)$ | $= \begin{cases} 1, & \text{if a truck leaves demand node } i \in D \text{ for the first time and travels on edge } (i,j) \in E \text{ on a route originating from depot } k \in W \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $a_{ijk}(\xi)$ | $= \begin{cases} 1, & \text{if a truck leaves demand node } i \in D \text{ for the last time and travels on edge } (i,j) \in E \text{ on a route originating from depot node } k \in W \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $b_{ijl}(\xi)$ | $= \begin{cases} 1, & \text{if a truck travels on edge } (j,l) \in E \text{ right after edge } (i,j) \in E, \text{ and if it does not leave demand node } j \in D \text{ for the first time and flow amount on edge } (i,j) \in E \text{ is equal to the flow amount of edge } (j,l) \in E \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $z_{ik}(\xi)$ | $= \begin{cases} 1, & \text{if a drone visits demand node } i \in D \text{ from depot node } k \in W_i \text{ under realisation } \xi \in \Omega \\ 0, & \text{otherwise} \end{cases}$ |
| $f_{ij}(\xi)$ | Commodity flow on edge $(i,j) \in E'$ under realisation $\xi \in \Omega$ |
| $p_i(\xi)$ | Drone arrival time at demand node $i \in D$ under realisation $\xi \in \Omega$ |
| $\phi_i(\xi)$ | Maximum flow into demand node $i \in D$ under realisation $\xi \in \Omega$ |
| $\psi_i(\xi)$ | Minimum flow into demand node $i \in D$ under realisation $\xi \in \Omega$ |
| $\sigma_i(\xi)$ | Minimum flow out of demand node $i \in D$ under realisation $\xi \in \Omega$ |
| $u_{ik}(\xi)$ | Number of times demand node $i \in D$ is visited by truck originating from depot node $k \in W$ under realisation $\xi \in \Omega$ |
| $h_{ik}(\xi)$ | Number of times demand node $i \in D$ is visited after the first visit by truck originating from depot node $k \in W$ under realisation $\xi \in \Omega$ |

(e.g., Caunhye et al. 2016, Hsueh et al. 2008, Mete and Zabinsky 2010), on the other hand, imply an increase in the time required to traverse a road segment, should it be disrupted. In this paper, we take the latter approach and make use of a set of discrete disaster scenarios. In each scenario, we consider different travel times due to the effects of the disaster on each road segment.

### 3.3. A two-stage stochastic mixed integer programming model

We formulate the problem as a two-stage stochastic mixed integer program with an objective of minimising the expected average arrival time for timely delivery. The first-stage decisions involve the locations of depots and the second-stage decisions relate to the routing of trucks and sequencing of drone visits. We list relevant notation in Table 1.

We model the problem on an undirected graph $G = (N, E)$ where vertices in $N$ are demand locations (denoted by set $D$), and candidate depot locations (represented by set $W$). Edges in $E$ correspond to direct road connections between any two locations in $N$. The pre-determined number of depots to open is denoted by $n$ and decision variables $x_k$ indicate whether a candidate location $k \in W$ is deployed as a depot.

To minimise the average arrival time of a drone from a depot $k \in W$ to a given set of demand nodes, the visiting order should follow the non-decreasing order of the drone's round trip flight time to demand nodes (Blazewicz et al., 2019). The set $D_k$ denotes the ordered list of demand nodes that a drone can visit and return to the candidate depot $k \in W$ within its endurance, arranged in non-decreasing order of travel times. We define the set $W_i$ as the depots $k \in W$ that are within the endurance limit for a return trip to the demand node $i \in D$.

We model uncertainty in travel times on the road network using discrete scenarios $\xi \in \Omega$. In this context, the stochastic parameters $t_{ij}(\xi)$ specify the truck travel time on edge $(i,j) \in E$ for each scenario $\xi \in \Omega$.

The decision variables $y_{ijk}(\xi)$ denote whether the truck departing from a depot node $k \in W$ travels on edge $(i,j) \in E$, under scenario $\xi \in \Omega$. Each demand node is assigned to a truck or drone dispatched from a depot $k \in W$ under scenario $\xi \in \Omega$, indicated by decision variables $c_{ik}(\xi)$ and $z_{ik}(\xi)$, respectively. Drone arrival time at demand node $i \in D$ under scenario $\xi \in \Omega$ is given by $p_i(\xi)$.

We use a commodity flow formulation to model multiple truck visits to demand nodes in a non-complete network (Letchford et al., 2013). To do so, we add a dummy sink node $s$ to the node set to obtain the augmented set $N'$, and create dummy edges connecting all demand nodes to $s$ to form the expanded edge set $E'$. The commodity flow variable $f_{ij}(\xi)$ carries demand units on edge $(i,j) \in E'$ to satisfy the remaining demands on the truck tour under scenario $\xi \in \Omega$. The truck delivers one unit in each first visit to a demand node, as well as one unit of flow to $s$. From a modelling perspective, flow starts from an open depot and after demand nodes are visited, and concludes at the sink node. However, it is important to note that the return of the truck to the depot is not a concern, as the objective function minimises the expected average arrival time to demand nodes. Although tours could be modelled to terminate at the depots and this would still be valid, it would require constructing the return path, which could result in unnecessarily lengthy tours. Therefore, using a single dummy node to terminate all truck tours is a modelling choice made for simplification purposes.

We exemplify the expansion of the network and values of commodity flow variables on a simple network in Fig. 1, where Fig. 1(a) shows the original network and Fig. 1(b) depicts the expanded network and an example truck tour that follows the route 1-2-3-2-4-$s$. Commodity flow amounts are provided for each corresponding flow variable. The initial flow amount, $f_{1,2}(\xi)$, is set to 4, reflecting the total needs of three demand nodes and a sink node. Flow amounts decrease by one unit on the first visits. However, when Node 2 is visited a second time, the incoming and the outgoing flow amounts become equal, such that $f_{3,2}(\xi) = f_{2,4}(\xi) = 2$.

The definition of commodity flow variables also allows us to calculate the average arrival time to nodes visited by trucks. Using same route on Fig. 1(a), the arrival times at nodes 2, 3, and 4 are calculated as $t_{1,2}(\xi)$, $t_{1,2}(\xi)+t_{2,3}(\xi)$, and $t_{1,2}(\xi)+t_{2,3}(\xi)+t_{3,2}(\xi)+t_{2,4}(\xi)$, respectively. This yields the summation of arrival times as $3t_{1,2}(\xi)+2t_{2,3}(\xi)+t_{3,2}(\xi)+t_{2,4}(\xi)$, which is equivalent to $f_{1,2}(\xi)t_{1,2}(\xi)+f_{2,3}(\xi)t_{2,3}(\xi)+f_{3,2}(\xi)t_{3,2}(\xi)+f_{2,4}(\xi)t_{2,4}(\xi)-[t_{1,2}(\xi)+t_{2,3}(\xi)+t_{3,2}(\xi)+t_{2,4}(\xi)]$. More generally, the total arrival time to truck-visited nodes can be written as:

$$\sum_{(i,j)\in E} f_{ij}(\xi)t_{ij}(\xi) - \sum_{k\in W}\sum_{(i,j)\in E} y_{ijk}(\xi)t_{ij}(\xi).$$

We define additional decision variables to ensure the commodity flows follow the aforementioned structure. Binary decision variables $w_{ijk}(\xi)$ indicate the truck's first departure from node $i$ when it travels on edge $(i,j) \in E$. Decision variables $a_{ijk}(\xi)$ mark the last departure of the truck from node $i \in D$ in the same way. Binary decision variables $b_{ijl}(\xi)$ indicate whether the truck's traversal of edge $(i,j) \in E$ is immediately followed by that of edge $(j,l) \in E$ and this is not the truck's first departure from node $j \in D$. The number of times a demand node $i \in D$ is visited by a truck from a depot node $k \in W$ is denoted by $u_{ik}(\xi)$ and the number of additional visits to a demand node after its demand is satisfied is indicated by $h_{ik}(\xi)$. For example, Node 3 in Fig. 1(b) is visited once, so the first and the last departures happen on the same edge, leading to $w_{3,2,1}(\xi) = a_{3,2,1}(\xi) = 1$. However, Node 2 is visited twice with first and last departures occurring on different edges, resulting in $w_{2,3,1}(\xi) = a_{2,4,1}(\xi) = 1$. Since only Node 2 is visited more than once, the only decision variable $b_{ijl}(\xi)$ that takes a positive value is $b_{3,2,4}(\xi) = 1$, indicating the immediate route 3-2-4 on the second departure from Node 2. Furthermore, Nodes 3 and 4 are visited once, hence $u_{3,1}(\xi) = u_{4,1}(\xi) = 1$, whereas $h_{3,1}(\xi) = h_{4,1}(\xi) = 0$. Similarly, $u_{2,1}(\xi) = 2$ and $h_{2,1}(\xi) = 1$ as Node 2 is visited twice.

We also denote the decision variables representing the maximum flow into, minimum flow into, and minimum flow out of demand node $i \in D$ by $\phi_i(\xi)$, $\psi_i(\xi)$, and $\sigma_i(\xi)$, respectively. These decision variables are required to assign the commodity flow amounts on edges to ensure that the minimum flow out of a demand node happens at the time of the last departure from that node and one unit of maximum incoming flow into a node is left after the first departure from that node.

Based on the described sets, parameters and decision variables, we formulate the problem as a two-stage stochastic mixed integer program as follows:

$$\min \quad E_{\xi|x}[g(\xi)] \tag{1}$$

$$\text{s.t.} \quad \sum_{k\in W} x_k \le n \tag{2}$$

$$x_k \in \{0,1\} \quad \forall k \in W \tag{3}$$

where $g(\xi)$ is the optimal solution to the following second-stage model:

$$\min \quad \frac{\sum_{(i,j)\in E} f_{ij}(\xi)t_{ij}(\xi) - \sum_{k\in W}\sum_{(i,j)\in E} y_{ijk}(\xi)t_{ij}(\xi) + \sum_{i\in D} p_i(\xi)}{|D|} \tag{4}$$

$$\text{s.t.} \quad p_{i_m}(\xi) \ge \sum_{j<m} z_{i_j k}(\xi)v_{i_j k} + \frac{z_{i_m k}(\xi)v_{i_m k}}{2} - M(1 - z_{i_m k}(\xi)) \qquad \forall k \in W,\ \forall i_m \in D_k \tag{5}$$

$$\sum_{k\in W} c_{jk}(\xi) + \sum_{k\in W_j} z_{jk}(\xi) = 1 \qquad \forall j \in D \tag{6}$$

$$\sum_{\substack{i\in N\\ i\neq j}} (y_{ijk}(\xi) + y_{jik}(\xi)) \le 2|D|c_{jk}(\xi) \qquad \forall j \in D,\ \forall k \in W \tag{7}$$

$$z_{ik}(\xi) \le x_k \qquad \forall i \in D,\ \forall k \in W_i \tag{8}$$

$$y_{ijk}(\xi) \le x_k \qquad \forall (i,j) \in E,\ \forall k \in W \tag{9}$$

$$\sum_{\substack{i\in D\\ i\neq k}} y_{kik}(\xi) \le x_k \qquad \forall k \in W \tag{10}$$

$$\sum_{\substack{j\in W\\ j\neq k}}\sum_{\substack{i\in N\\ i\neq k}} y_{kij}(\xi) = 0 \qquad \forall k \in W \tag{11}$$

$$\sum_{\substack{j\in N\\ j\neq i}} y_{ijk}(\xi) = \sum_{\substack{j\in N\\ j\neq i}} y_{jik}(\xi) \qquad \forall i \in N,\ \forall k \in W \tag{12}$$

$$\sum_{\substack{j\in N\\ j\neq i}} y_{jik}(\xi) = u_{ik}(\xi) \qquad \forall i \in D,\ \forall k \in W \tag{13}$$

$$h_{ik}(\xi) \ge u_{ik}(\xi) - 1 \qquad \forall i \in D,\ \forall k \in W \tag{14}$$

$$\sum_{\substack{i\in N\\ i\neq k}} f_{ki}(\xi) = \sum_{(i,j)\in E} y_{ijk}(\xi) - \sum_{i\in D} h_{ik}(\xi) \qquad \forall k \in W \tag{15}$$

$$\sum_{\substack{j\in N\\ j\neq i}} f_{ji}(\xi) - \sum_{j\in N'} f_{ij}(\xi) = 1 - \sum_{k\in W} z_{ik}(\xi) \qquad \forall i \in D \tag{16}$$

$$\sum_{i\in N} f_{is}(\xi) = \sum_{k\in W} x_k \tag{17}$$

$$f_{ij}(\xi) \le (|D|+1)\sum_{k\in W} y_{ijk}(\xi) \qquad \forall (i,j) \in E \tag{18}$$

$$f_{ij}(\xi) \ge y_{ijk}(\xi) \qquad \forall j \in D,\ \forall (i,j) \in E,\ \forall k \in W \tag{19}$$

$$f_{is}(\xi) \le \sum_{\substack{k\in W\\ k\neq i}} y_{ikk}(\xi) \qquad \forall i \in N \tag{20}$$

$$w_{ijk}(\xi) \le y_{ijk}(\xi) \qquad \forall i \in D,\ \forall (i,j) \in E,\ \forall k \in W \tag{21}$$

$$a_{ijk}(\xi) \le y_{ijk}(\xi) \qquad \forall i \in D,\ \forall (i,j) \in E,\ \forall k \in W \tag{22}$$

(a) Original network (b) Modified network

**Fig. 1.** Illustrative example for expansion of the network and values of commodity flow variables.

$$\sum_{k\in W}\sum_{\substack{j\in N\\ j\neq i}} w_{ijk}(\xi) = 1 - \sum_{k\in W} z_{ik}(\xi) \qquad \forall i \in D \tag{23}$$

$$\sum_{k\in W}\sum_{\substack{j\in N\\ j\neq i}} a_{ijk}(\xi) = 1 - \sum_{k\in W} z_{ik}(\xi) \qquad \forall i \in D \tag{24}$$

$$\sigma_i(\xi) \geq f_{ij}(\xi) - (|D|+1)(1 - \sum_{k\in W} a_{ijk}(\xi)) \qquad \forall i,j \in D,\ \forall (i,j) \in E \tag{25}$$

$$\sigma_i(\xi) \geq f_{is}(\xi) - (|D|+1)(1 - \sum_{k\in W} a_{ikk}(\xi)) \qquad \forall i \in D \tag{26}$$

$$\phi_i(\xi) \geq f_{ji}(\xi) \qquad \forall i \in D,\ \forall (i,j) \in E \tag{27}$$

$$f_{ij}(\xi) \geq \phi_i(\xi) - 1 - (|D|+1)(1 - \sum_{k\in W} w_{ijk}(\xi)) \qquad \forall i,j \in D,\ \forall (i,j) \in E \tag{28}$$

$$f_{is}(\xi) \geq \phi_i(\xi) - 1 - (|D|+1)(1 - \sum_{k\in W} w_{ikk}(\xi)) \qquad \forall i \in D \tag{29}$$

$$\psi_i(\xi) \leq f_{ji}(\xi) + (|D|+1)(1 - \sum_{k\in W} y_{jik}(\xi)) \qquad \forall i \in D,\ \forall (i,j) \in E \tag{30}$$

$$\sigma_i(\xi) \leq \psi_i(\xi) \qquad \forall i \in D \tag{31}$$

$$\sum_{\substack{i\in N\\ i\neq j}} b_{ijl}(\xi) \leq \sum_{k\in W} y_{jlk}(\xi) \qquad \forall j \in D,\ \forall (i,j) \in E \tag{32}$$

$$\sum_{\substack{l\in N\\ l\neq j}} b_{ijl}(\xi) \leq \sum_{k\in W} y_{ijk}(\xi) \qquad \forall j \in D,\ \forall (i,j) \in E \tag{33}$$

$$\sum_{\substack{i\in N\\ i\neq j}} b_{ijl}(\xi) \geq \sum_{k\in W} (y_{jlk}(\xi) - w_{jlk}(\xi)) \qquad \forall j \in D,\ \forall (j,l) \in E \tag{34}$$

$$f_{jl}(\xi) \geq f_{ij}(\xi) - (|D|+1)(1 - b_{ijl}(\xi)) \qquad \forall j,l \in D,\ \forall (i,j),(j,l) \in E \tag{35}$$

$$f_{jl}(\xi) \leq f_{ij}(\xi) + (|D|+1)(1 - b_{ijl}(\xi)) \qquad \forall j,l \in D,\ \forall (i,j),(j,l) \in E \tag{36}$$

$$f_{js}(\xi) \geq f_{ij}(\xi) - (|D|+1)(1 - b_{ijk}(\xi)) \qquad \forall j \in D,\ \forall (i,j) \in E,\ \forall k \in W \tag{37}$$

$$f_{js}(\xi) \leq f_{ij}(\xi) + (|D|+1)(1 - b_{ijk}(\xi)) \qquad \forall j \in D,\ \forall (i,j) \in E,\ \forall k \in W \tag{38}$$

$$y_{ijk}(\xi),\ w_{ijk}(\xi),\ a_{ijk}(\xi) \in \{0,1\} \qquad \forall (i,j) \in E,\ \forall k \in W \tag{39}$$

$$b_{ijl}(\xi) \in \{0,1\} \qquad \forall j \in D,\ \forall (i,l) \in E \tag{40}$$

$$z_{ik}(\xi) \in \{0,1\} \qquad \forall i \in D,\ \forall k \in W_i \tag{41}$$

$$f_{ij}(\xi) \geq 0 \qquad \forall (i,j) \in E' \tag{42}$$

$$p_i(\xi),\ \phi_i(\xi),\ \psi_i(\xi),\ \sigma_i(\xi) \geq 0 \qquad \forall i \in D \tag{43}$$

$$u_{ik}(\xi),\ h_{ik}(\xi) \geq 0 \qquad \forall i \in D,\ \forall k \in W \tag{44}$$

The objective function (1) minimises the expected objective value of the second stage problem. Constraint (2) ensures that at most $n$ depots are opened. Constraints (3) define the domain of the first stage variables. Given the first stage decisions, the second stage problem objective (4) minimises the average arrival time to demand nodes under the realisation $\xi \in \Omega$. The first two terms in the numerator of function (4) calculate the total arrival time at nodes visited by trucks, while the final part of the numerator of function (4) sums the arrival times at nodes visited by drones. The total arrival time to all demand nodes is then divided by the number of demand nodes to obtain the average arrival time.

Constraints (5) are employed to calculate the arrival time at each demand node $i \in D$ visited by a drone, denoted by $p_i(\xi)$. As each drone is assumed to make a single delivery per trip and return to the depot before the next launch, the first part of the constraint sums the round-trip flight times to all nodes preceding $i$ in $D_k$, i.e., demand nodes $j$ for which the round trip from depot $k$ is no longer than that to $i$ that are visited from this depot. The second part of the constraint then adds the one-way flight time from the depot to node $i$, representing the moment the drone reaches the node.

Constraints (6) guarantee that each demand node is assigned either to a drone or a truck. Constraints (7) maintain route consistency by ensuring that a demand node $j \in D$ is included in the truck tour originating from depot $k \in W$ only if it is assigned to that truck. Constraints (8) limit drone flights to demand nodes from open depots that satisfy the endurance limit. However, the endurance limit is enforced for all drone trips through the domain of the decision variable $z_{ik}(\xi)$, as defined in Constraints (41). The variable domain utilises the set $W_i$ to specify the depots from which a drone can make a return trip to visit demand node $i$ within the endurance limit. Thus, by the variable definition, only endurance-feasible drone trips are permitted in the model.

Constraints (9) ensure that a truck can only travel on edge $(i,j) \in E$ on a route originating from an open depot $k \in W$. Meanwhile, constraints (10) ensure that only one truck can leave an open depot. Route consistency is maintained through constraints (11), which ensure that a truck tour starting from depot $k \in W$ does not originate from another depot. Constraints (12) focus on truck flow balance such that for each node, the number of incoming and outgoing truck flows are equal. Constraints (13) calculate the values of the $u_{ik}$ variables. By subtracting one unit from each $u_{ik}$ decision variable, constraints (14) provide a lower bound for decision variables $h_{ik}$, the number of

additional visits a truck made to a demand node $i \in D$ after its first visit.

Constraints (15) to (38) are commodity flow constraints. Constraints (15) ensure that the outgoing commodity flow from depot $k \in W$ is equal to the number of demand nodes the truck is scheduled to visit, plus an additional unit accounting for the sink node. If a demand node $i \in D$ is not visited by any drone, as indicated by the summation on the right-hand side of the constraint, it must be on a truck tour. Constraint (16) then ensures that one unit of flow is subtracted from the route after leaving node $i$. Thus, constraint (17) makes sure that the incoming flow amounts to the sink node equal to the number of open depots. By doing so, constraints (17) force trucks to make a tour from an open depot. Constraints (18) specify that if a truck does not traverse an edge, the associated commodity flow cannot be positive. Conversely, commodity flows must be positive when a truck traverses an arc, dictated by constraints (19). Constraints (20) link the commodity flow from node $i \in N$ to the sink node with the return of a truck from node $i \in N$ to a depot, synchronising the movement of both trucks and commodities. Constraints (21) and (22) prohibit decision variables $w_{ijk}(\xi)$ and $a_{ijk}(\xi)$ from taking positive values if the truck from depot $k \in W$ does not traverse the edge $(i,j) \in E$. Constraints (23) and (24) guarantee that if a demand node $i \in D$ is not assigned to any drone, it must be visited by a truck. In that case, exactly one of the $w_{ijk}(\xi)$ variables and exactly one of the $a_{ijk}(\xi)$ variables takes a positive value.

Constraints (25) and (26) define the minimum commodity flow outgoing from node $i \in D$ by utilising the $a_{ijk}(\xi)$ variables. These constraints stipulate that the minimum flow happens at the time of the last departure from that node. In Fig. 1(b), Node 2 is visited twice and the outgoing flow amounts from that node are three and two as seen on the network. Decision variable $a_{2,4,1}(\xi) = 1$ indicates that the latest departure from Node 2 is on the edge $(2,4)$, therefore, the minimum outgoing flow from Node 2 is bounded by the flow amount on the edge $(2,4)$. Constraints (27) assign the maximum flow into node $i$. Referring to the same example in Fig. 1(a), the maximum flow into Node 2 is set four as the incoming flow amounts are four and two. By using maximum flow into node $i \in D$, constraints (28) and (29) ensure that at the time of the first departure from node $i$, one unit is left at the node. On the illustrative example, outgoing flow amount after the initial visit to Node 2 becomes three on edge $(2,3)$ as $w_{2,3,1}$ embarks the first departure from that node. Constraints (30) put an upper limit to $\psi_i(\xi)$ variables representing the minimum flow into node $i \in D$. On the same example, minimum flow into Node 2 is bounded by two as the incoming flows are four and two. Constraints (31) ensure that for each demand node $i \in D$, the minimum flow outgoing from the node is less than or equal to the minimum flow incoming to it.

We prohibit the $b_{ijl}(\xi)$ variables from taking positive values unless the truck traverses these edges by constraints (32) and (33). Furthermore, if edge $(j,l) \in E$ is traversed and this traversal is not the truck's first departure from node $j \in D$, constraints (34) ensure that $b_{ijl}(\xi)$ variables must take positive values. On the illustrative example, since traversing the edge $(2,4)$ is not the first departure from Node 2, $w_{2,4,1} = 0$ and $b_{3,2,4}$ is forced to take value 1. Finally, constraints (35) to (38) guarantee that if $b_{ijl}(\xi)$ variables take positive value, flow on edge $(i,j) \in E$ is equal to the flow on the edge $(j,l) \in E$. In the example network, $b_{3,2,4} = 1$ shows the immediate tour marking the second departure from Node 2. Thus, the flow amounts on those edges set equal $f_{3,2} = f_{2,4} = 2$.

Constraints (39)–(44) define the domains of the decision variables.

## 4. A variable neighbourhood search heuristic

The problem defined in Section 3 can be easily shown to be $\mathcal{NP}$-hard, as the deterministic version with no drones and no location decisions is equivalent to the $\mathcal{NP}$-hard multi-depot $m$-TSP (e.g., Oberlin et al. 2009). Furthermore, it will become apparent in Section 5 that even very small instances of the problem cannot be solved to optimality by commercial solvers. Consequently, we propose a heuristic approach based on variable neighbourhood search to solve practical-sized instances of the problem, which is explained in detail in this section.

### 4.1. Overview of the heuristic algorithm

Our proposed heuristic approach consists of two phases. In the first phase, we perform a local search for depot selection. Subsequently, the second phase evaluates open depot sets identified in the local search by constructing the truck routes and drone schedules. This phase consists of the initial tour construction and variable neighbourhood search-based improvement steps. Additional notation defined for the heuristic algorithm is provided in Table 2.

Algorithm 1 summarises our overall approach, which starts with a random open depot set. Here, $N_{depot}$ defines a swap neighbourhood, where we close one of the open depots and open one that is currently closed. We evaluate all neighbouring open depot sets (in lines 6 to 11) and we move to the set that yields the best-expected latency. Each new set of open depots is evaluated in the second phase, which is summarised in the next section.

**Algorithm 1** Overview of the heuristic algorithm

**Function** StochasticLocationRouting($N_{depot}, \Omega, k_{max}, l_{max}, i_{max}$)
1: Initialise $X$ randomly
2: $f_{min} \leftarrow DepotEvaluation(X, \Omega, k_{max}, l_{max}, i_{max})$
3: $f_{new} \leftarrow f_{min}$
4: $StopFlag \leftarrow 0$
5: **while** $StopFlag = 0$ **do**
6: **for** all $Y \in N_{depot}(X)$ **do**
7: **if** $DepotEvaluation(Y, \Omega, k_{max}, l_{max}, i_{max}) < f_{new}$ **then**
8: $X' \leftarrow Y$
9: $f_{new} \leftarrow DepotEvaluation(Y, \Omega, k_{max}, l_{max}, i_{max})$
10: **end if**
11: **end for**
12: **if** $f_{new} = f_{min}$ **then**
13: $StopFlag \leftarrow 1$
14: **else**
15: $X \leftarrow X'$
16: $f_{min} \leftarrow f_{new}$
17: **end if**
18: **end while** ▷ Until no improving set of open depots is found
**return** $X'$

### 4.2. Depot evaluation heuristic

The second phase of the algorithm is given in Algorithm 2. For a given open depot set, this phase constructs the truck tours and drone schedules for each distinct scenario and calculates the expected objective over all scenarios.

**Algorithm 2** Depot evaluation heuristic

**Function** DepotEvaluation($X, \Omega, k_{max}, l_{max}, i_{max}$)
**for** all $\xi$ **in** $\Omega$ **do**
$R(X,\xi) \leftarrow InitialConstruction(X,\xi)$
$R(X,\xi) \leftarrow Improvement(R(X,\xi), k_{max}, l_{max}, i_{max})$
**end for**
**return** $f(X) \leftarrow E_{\xi|X}[h(R(X,\xi))]$ ▷ Expected summation of arrival times

Under a specific scenario, the procedure involves two key stages: initial construction and subsequent improvement. Initial construction finds preliminary truck tours and drone schedules, which may not be immediately feasible, with an anticipation of reaching feasible routes during the improvement steps.

**Table 2**
Additional notation used for the heuristic.

| | |
|---|---|
| *Index Sets* | |
| $X$ | Open depots |
| $U_k$ | Unvisited demand nodes assigned to depot $k \in W$ |
| $TT_k$ | Demand nodes in order of truck visit starting from depot $k \in W$ |
| $DL_k$ | Demand nodes assigned to a drone dispatched from depot $k \in W$ |
| $DS_k$ | Demand nodes in order of drone visit starting from depot $k \in W$ |
| *Algorithm Parameters* | |
| $k_{max}$ | Number of neighbourhoods defined for the shake step |
| $l_{max}$ | Number of neighbourhoods defined for the local search |
| $i_{max}$ | Maximum number of iterations |
| *Evaluation Functions* | |
| $h(R(X,\xi))$ | Evaluation of routes under depot set $X$ and scenario $\xi \in \Omega$, i.e., sum of arrival times at all demand nodes given the scenario and open depots |

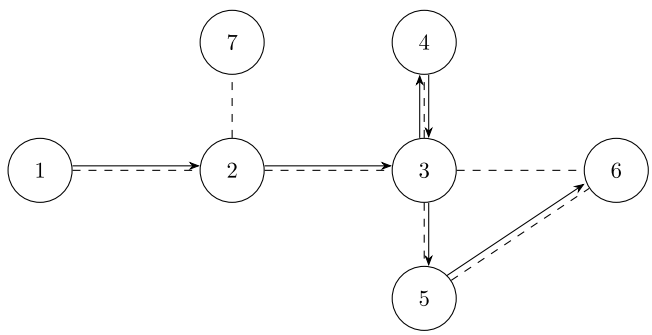



**Fig. 2.** Illustrative example for truck backtracking.

#### 4.2.1. Route construction by a greedy heuristic approach

Algorithm 3 provides the initial construction process. It starts by assigning demand nodes to the closest open depot based on direct geodesic distance. For each open depot, we construct a truck tour by modifying the nearest neighbour heuristic for the Travelling Salesman Problem. The initial construction heuristic aims to construct a truck tour to visit all allocated demand nodes with minimum backtracking, assuming no drone is available. It adds the nearest assigned unvisited node to the truck tour according to the truck travel times of the scenario under consideration. The process involves continuously inserting the closest unvisited nodes (lines 17 to 20).

If there is no direct road link to an unvisited node (checked in lines 10 to 16), the truck backtracks to a previously visited node to explore alternative ways to visit the remaining nodes (in lines 21 to 44). However, a truck can face a dead-end and may not find a backtrack node to return to in the initial tour construction step. In that case, we assign the remaining unvisited demand nodes to the drone of the same depot (in lines 38 to 41).

After a backtrack node is found, instead of retracing the exact truck tour, the shortest backtracking route is used. We illustrate the backtracking of a truck on a small network in Fig. 2. Consider the current truck tour $1 \to 2 \to 3 \to 4 \to 3 \to 5 \to 6$, shown in solid arrows. The only unvisited node is 7, which does not share an edge with the current node, 6. By progressively evaluating each node on the current tour backward, node 2 is found as the point of backtrack because it shares an edge with the unvisited node 7. Following Algorithm 4, we find the backtracking route. Step 1 extracts the partial tour as $2 \to 3 \to 4 \to 3 \to 5$. Step 2 simplifies the partial tour by removing the redundant visits, resulting in $2 \to 3 \to 5$. Step 3 progressively seeks a shared edge between the current node and a node on the simplified tour. There is no edge between node 2 and node 6, but there is one between node 3 and node 6. Thus, from the current node 6 to node 3, the truck backtracks to node 2 and visits node 7. As a result, the final truck tour becomes $1 \to 2 \to 3 \to 4 \to 3 \to 5 \to 6 \to 3 \to 2 \to 7$.

#### 4.2.2. Route improvement by variable neighbourhood search

When the initial truck tours and drone schedules are determined, we make use of a variable neighbourhood search (VNS) heuristic to improve the truck and drone routes. Algorithm 5 summarises our VNS procedure, which is run for a total of $i_{max}$ iterations. Throughout the VNS, we use the following neighbourhood structures, which refer to the set of all possible solutions that can be obtained by the given move applied on a current solution $r$:

$N_1(r)$: *truck to truck reassignment.* Move a demand node from a truck tour to a different truck tour.

$N_2(r)$: *truck to drone reassignment — same depot.* Remove a demand node from a truck tour and assign it to the drone dispatched from the same depot.

$N_3(r)$: *truck to drone reassignment — different depot.* Remove a demand node from a truck tour and assign it to a drone dispatched from a different depot.

$N_4(r)$: *drone to truck reassignment — same depot.* Reassign a drone-visited demand node to a truck tour originating from the same depot.

$N_5(r)$: *drone to truck reassignment — different depot.* Reassign a drone-visited demand node to a truck tour originating from a different depot.

$N_6(r)$: *drone to drone reassignment.* Reassign a drone-visited demand node to a different drone.

$N_7(r)$: *complete reconstruction of a truck tour.* Randomly select a truck and reconstruct its route by changing its initially visited demand node. Any demand node on the previous truck tour are assigned to the closest drone.

Each iteration of the algorithm starts with a shake move, where we randomly select a neighbouring solution from the current solution (in lines 5 to 6). Then, local search is performed from that solution (in lines 8 to 19). If the local search results in a better solution, we update the current solution and set the first neighbourhood for shaking. If however, local search does not provide a better solution, the next neighbourhood is set for shaking (in lines 20 to 25). This process continues until all shake neighbourhoods are implemented.

In the local search, we use a best improvement approach. This process is repeated for each neighbourhood until no further improvements can be achieved with the same neighbourhood structure (in lines 9 to 12). If no improving solution is obtained from a particular neighbouring structure, we move to the next neighbourhood. If, however, we obtain an improving solution, we return to the first neighbourhood structure (in lines 13 to 18). Neighbouring solutions are explored in order of the above list as preliminary results suggest that no particular order of neighbourhoods consistently outperforms others.

We use the same neighbourhood structures for both steps, with the exception of $N_7$, which is only used in shaking. In the set of neighbouring solutions, we consider all feasible neighbouring routes and infeasible ones violating the drone flight duration. Infeasible drone assignments are included in the search with a large dummy flight duration to make them undesirable.

**Algorithm 3** Initial truck tour and drone schedule construction heuristic

**Function** InitialConstruction$(X, \xi)$
$U_k, TT_k, DL_k, DS_k \leftarrow \{\}$
**for** all $i \in D$ **do**
  Set $k$ as the closest open depot to demand node $i$
  $U_k \leftarrow U_k \cup \{i\}$ ▷ Assign node to unvisited nodes list of depot
**end for**
**for** all $k \in X$ **do**
  $current \leftarrow k$ ▷ Assign depot as current node
  $TT_k \leftarrow TT_k \cup \{current\}$
  **while** $|U_k| > 0$ **do**
    $immediateConnection \leftarrow False$
    **for** all $j \in U_k$ **do**
      **if** $(current, j) \in E$ **then**
        $immediateConnection \leftarrow True$
        **Break** ▷ There is at least one unvisited node truck can immediately visit
      **end if**
    **end for**
    **if** $immediateConnection$ **then**
      $node \leftarrow \arg\min_{j \in U_k, (current,j) \in E} \{t_{current,j}(\xi)\}$
      Move $node$ from $U_k$ to $TT_k$
      $current \leftarrow node$
    **else** ▷ Find a node to backtrack
      $T \leftarrow TT_k \setminus \{\text{last element in } TT_k\}$ ▷ Copy truck tour until the current node
      $StopFlag \leftarrow 0$
      **while** $|T| > 0$ **do**
        $BackNode \leftarrow$ last element in $T$
        **for** all $j \in U_k$ **do**
          **if** $(BackNode, j) \in E$ **then**
            Apply Algorithm 4 and find the backtracking route $B_k$
            $TT_k \leftarrow TT_k \cup B_k$
            $node \leftarrow \arg\min_{j \in U_k, (BackNode,j) \in E} \{t_{BackNode,j}(\xi)\}$
            Move $node$ from $U_k$ to $TT_k$
            Set $current \leftarrow node$ and $StopFlag \leftarrow 1$
            **Break** ▷ Truck can visit unvisited node if it backtracks to $BackNode$
          **end if**
        **end for**
        **if** $StopFlag = 0$ **then**
          $T \leftarrow T \setminus \{BackNode\}$
          **if** $|T| = 1$ **then** ▷ Truck faces a dead-end and no backtrack node is found
            $DL_k \leftarrow U_k$ ▷ Assign all unvisited nodes to drone list
            $U_k \leftarrow \{\}$
          **end if**
        **end if**
      **end while**
    **end if**
  **end while**
  **for** all $j \in D_k$ **do**
    **if** $j \in DL_k$ **then**
      $DS_k \leftarrow DS_k \cup \{j\}$ ▷ Construct drone schedule
    **end if**
  **end for**
  **for** all $j \in DL_k$ **do**
    **if** $j \notin D_k$ **then**
      $DS_k \leftarrow DS_k \cup \{j\}$ ▷ Add infeasible drone assignments to the drone schedule
    **end if**
  **end for**
**end for**
**return** $R(X, \xi)$

**Algorithm 4** Process of finding the shortest backtracking route

**Step 1. Extracting the partial tour**
Extract the exact tour from the point of backtrack until the current node.

**Step 2. Simplifying the partial tour**
**if** there are revisited nodes **then**
  **while** there are revisited nodes **do**
    Remove the tour segment between the first and the last visit to the revisited node.
  **end while**
**end if**

**Step 3. Finding the shortest route**
**for** each node in the simplified tour **do**
  **if** there is direct connection between the node and current node **then**
    Truck continues to the node and follows the simplified tour backwards until reaching the backtrack node.
    **Break for**
  **end if**
**end for**

Truck follows the simplified tour backwards from the current node until reaching the point of backtrack.

**Algorithm 5** Truck tour and drone schedule improvement heuristic

**Function** Improvement$(R, k_{max}, l_{max}, i_{max})$
$i \leftarrow 0$
**while** $i \leq i_{max}$ **do**
  $k \leftarrow 1$ ▷ Initialise shake neighbourhood
  **while** $k < k_{max}$ **do**
    $w \leftarrow \lfloor 1 + rand(0,1) \times |N_k(R)| \rfloor$
    $R' \leftarrow N_k(R)[w]$ ▷ Updated to $w^{th}$ element of the neighbourhood
    $l \leftarrow 1$ ▷ Initialise local search neighbourhood
    **while** $l \leq l_{max}$ **do**
      **while** $h(R') < h(R'')$ **do**
        $R'' \leftarrow R'$
        $R' \leftarrow \arg\min_{R \in N_l(R'')} \{h(R)\}$
      **end while**
      **if** $f(R'') \leq h(R')$ **then**
        $R' \leftarrow R''$
        $l \leftarrow 1$
      **else**
        $l \leftarrow l + 1$
      **end if**
    **end while** ▷ Until all local search neighbourhoods are used
    **if** $f(R') \leq h(R)$ **then**
      $R \leftarrow R'$
      $k \leftarrow 1$
    **else**
      $k \leftarrow k + 1$
    **end if**
  **end while** ▷ Until all shake neighbourhoods are used
  $i \leftarrow i + 1$
**end while** ▷ Until maximum iteration is reached
**return** $R$

## 5. Computational experiments

In this section, we evaluate the performance of our solution approach by comparing the results of our heuristic with those obtained from Gurobi on small-sized instances. We also apply the heuristic to a case study derived from the 2011 Van Earthquake in Turkiye to draw managerial insights, perform sensitivity analysis, and assess the value of incorporating stochasticity into our approach. In the first part of this section, we explain the problem instances used in the computational experiments. Then, we provide the results of our heuristic comparison with Gurobi, and those of the case study based on the Van Earthquake.

The mathematical models in our experiments are solved using Gurobi 11.0.3 through the Python API. The heuristic algorithm is implemented in C++ using Visual Studio 2022. All computational experiments are performed on a Windows 10 PC equipped with an Intel(R) Core(TM) i7-10510U CPU @ 2.30 GHz and 16 GB of RAM.

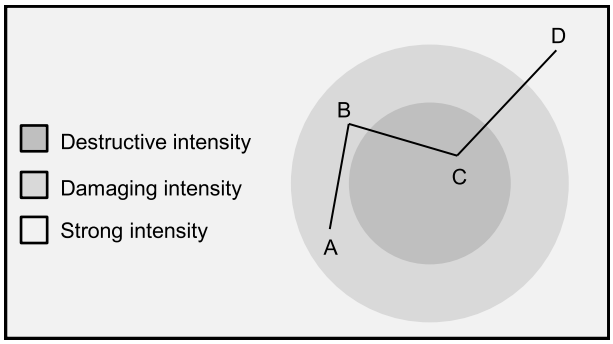



**Fig. 3.** Damage intensity areas around disaster epicentre.

### 5.1. Problem instances

In the following sections, we explain how we generate the small-sized instances and develop the case study.

#### 5.1.1. Small-sized instances

For our first set of experiments, we generate 10 instances consisting of 15 demand nodes and 5 candidate depots. For these instances, we locate two depots and consider 10 different disaster scenarios.

Demand nodes were randomly placed in a $5,000 \times 5,000$ unit square, with Euclidean distances and equal speeds for both vehicle types, ignoring drone travel time limits for simplicity. Each node is connected to each of its two closest neighbours with a probability of 0.7 each. In case no connection is formed in this way, the node is connected to its nearest neighbour. It is also connected to its third, fourth, and fifth closest neighbours with a probability of 0.1 for each. Candidate depot nodes are selected randomly from the demand nodes.

Disaster epicentres are randomly located, with two circular regions around them as described by Noyan et al. (2016). The radii of one-tenth and one-fifth of the square's side length around the epicentre, adjusted by a random multiplier U(0.8,1.6) to account for varying-sized disasters, define destructive and damaging regions, respectively. The remaining area is considered to have strong intensity. Damage intensity of each edge is determined by the highest intensity of its two end nodes, depending on their location. Travel times are inflated by random factors: U(1.6,2.5) for destructive, U(1.25,1.6) for damaging, and U(1.0,1.25) for strong intensities. An illustration for this is given in Fig. 3. Here, the end nodes of edge AB are both in the damaging region, so it is marked as having damaging intensity. On the other hand, edges BC and CD are classified under destructive intensity, as C is situated in the destructive zone. All small-sized instances used in this study are publicly available at https://github.com/hannantureciisik/parallel-truck-drone-small-instances.

#### 5.1.2. Case study instances

We use a case study based on the earthquake which occurred in Van, Turkiye on October 23, 2011, at a magnitude of 7.2, followed by an aftershock of magnitude 5.6 on November 9, 2011. These resulted in more than 600 people losing their lives and 2500 injured or hospitalised (IFRC, 2012).

We obtain the coordinates of the 94 demand locations from Noyan et al. (2016) and the truck travel times from Google Maps (Google, 2024a). We create symmetric truck travel times by taking the maximum of the travel times between each pair of locations in both directions. For drone travel times, we calculate the geodesic distances between locations first by using the 'geodesic' function in the 'geopy' library in Python and divide the distances by drone speed. We double the resulting time to obtain the return flight time. Following (Murray & Chu, 2015), we use 25 mph (approximately 40 km/h) as drone speed and 30 min as drone endurance. We elaborate further on additional values in the sensitivity analysis.

We generate the road network as follows: The driving path between each pair of demand points are found by using Google Maps Directions API (Google, 2024b). If any of the intermediate points on the path between a pair of demand locations is within 0.01 units of latitude and longitude difference of any other demand location, we do not form an immediate edge between the given pair. Otherwise, we create a direct edge between these two demand locations if their geodesic distance is less than 10 km.

We generate 10 case study instances with 15 candidate depot locations, selected as a subset of the demand nodes. The candidate locations are chosen such that none of them share an edge with each other. We aim to locate 5 depots, considering 50 disaster scenarios.

To generate representative earthquake scenarios, we select six locations along the Van and Yenikosk fault lines. These are the two active fault lines lying between Van Lake and Ercek Lake in the area (MTA, 2024) (see Online Appendix for the map representation). For each scenario, we randomly choose a location and generate the epicentre coordinates within a 5 km radius. To account for disasters of different sizes, we define two radii as a basis, 5 km and 10 km, to represent the same areas and increment these radii using a multiplier from a uniform distribution U(0.8,3). Additionally, we adjust truck travel times based on the edge classification. Edges within the highest destructive region based on their endpoints have travel times increased by multipliers from U(3,5), while those in damaging and strong categories see increases by multipliers from U(1.5,3) and U(1,1.5), respectively.

### 5.2. Computational results for smaller instances

Our smaller instances serve the purpose of evaluating the quality of heuristic solutions by comparison to the commercial solver Gurobi, for which we impose a 3-hour computational time limit. However, given the heavy computational burden to solve the two-stage stochastic mixed integer programming model, Gurobi is unable to close the MIP gap within the time limit to find the optimal solutions. To obtain the optimal solutions, we also run Gurobi by enumerating all possible sets of open depots and solving the resulting problem for each combination, where the open depot set with the minimum expected objective value provides the optimal solution.

In Table 3, we report the results of our experiments on smaller instances. These include the open depots, the integrality gap (denoted by % MIPGap), and the optimality gap (denoted by % Gap) of the Gurobi solutions after the 3-hour limit; the open depots and total CPU times (in seconds) for the Gurobi runs for the routing problem that yield the optimal solutions; and open depots, % Gap values, and CPU times for the heuristic with and without VNS improvement.

From Table 3, we observe that the heuristic consistently opens the optimal set of depots. Moreover, the average optimality gap of 0.3% indicates that the heuristic finds near-optimal truck routes and drone schedules under all scenarios. In five out of ten problem instances, the heuristic also finds the optimal truck–drone routes under all scenarios. The highest optimality gap among the remaining instances is 1.1%.

**Table 3**
Set of open depots, optimality gaps (where applicable), and CPU times (in seconds) of Gurobi results and those of the heuristic with and without VNS improvement.

| Instance | Gurobi after 3-hour limit | | | Gurobi for routing | | Heuristic - with improvement | | | Heuristic - without improvement | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Open Depots | % MIPGap | % Gap | Open Depots | CPU time[a] | Open Depots | % Gap | CPU time | Open Depots | % Gap | CPU time |
| 1 | {0,2} | 32.0 | < 0.1 | {0,2} | 475.2 | {0,2} | 0.9 | 7.4 | {2,3} | 43.3 | 0.1 |
| 2 | {0,2} | 35.9 | 8.8 | {0,1} | 409.2 | {0,1} | 1.1 | 8.5 | {3,4} | 36.8 | 0.2 |
| 3 | {3,4} | 38.8 | 3.6 | {0,3} | 135.9 | {0,3} | 0.5 | 4.6 | {3,4} | 27.3 | 0.1 |
| 4 | {0,1} | 35.2 | 6.3 | {1,2} | 1,058.2 | {1,2} | 0.0 | 10.2 | {2,4} | 71.1 | 0.1 |
| 5 | {2,4} | 27.4 | < 0.1 | {2,4} | 231.2 | {2,4} | 0.0 | 7.9 | {1,4} | 25.5 | 0.1 |
| 6 | {0,1} | 43.5 | 0.5 | {0,1} | 1,303.4 | {0,1} | 0.3 | 6.7 | {1,3} | 205.3 | 0.1 |
| 7 | {1,2} | 39.2 | 9.3 | {1,3} | 1,050.1 | {1,3} | 0.0 | 7.5 | {0,1} | 70.3 | 0.1 |
| 8 | {0,3} | 35.7 | 6.5 | {0,4} | 828.4 | {0,4} | 0.0 | 9.8 | {2,3} | 51.6 | 0.1 |
| 9 | {1,3} | 29.9 | 5.0 | {2,4} | 322.9 | {2,4} | 0.5 | 5.2 | {2,4} | 15.5 | 0.1 |
| 10 | {0,3} | 44.1 | 11.4 | {2,4} | 2,093.2 | {2,4} | 0.0 | 7.9 | {1,4} | 39.3 | 0.1 |
| Average | | **36.2** | **5.1** | | **790.8** | | **0.3** | **7.6** | | **58.6** | **0.1** |

[a] Total CPU time to enumerate all open depot combinations.

**Table 4**
Averages and 95% confidence intervals for percent increase in objective value and percent decrease in CPU time when each neighbourhood is excluded.

| | | $N_1$ | $N_2$ | $N_3$ | $N_4$ | $N_5$ | $N_6$ | $N_7$ |
|---|---|---|---|---|---|---|---|---|
| % increase in objective value | Avg | 0.66 | 0.62 | 0.71 | 3.24 | 1.03 | 0.83 | 5.66 |
| | 95% CI | (0.19, 1.13) | (0.28, 0.96) | (0.08, 1.34) | (1.67, 4.81) | (0.36, 1.70) | (0.32, 1.34) | (3.17, 8.15) |
| % decrease in CPU time | Avg | 0.26 | −1.05 | 4.87 | 6.84 | 0.13 | −3.29 | 5.66 |
| | 95% CI | (−14.45, 14.98) | (−16.30, 14.20) | (−6.86, 16.60) | (0.71, 12.97) | (−10.60, 10.86) | (−18.05, 11.48) | (−2.79, 14.11) |

The small interval of the optimality gap suggests the heuristic is robust in generating high-quality solutions. Furthermore, while the heuristic solves the instances within an average CPU time of only 3.6 s, We also observe that Gurobi cannot reduce the MIP Gap to less than 27.4% in any of the instances within 3-hour limit and takes an average of 790.8 s to solve the routing problem for all possible sets of open depots. In all, these underline the efficiency and effectiveness of the proposed heuristic, as it not only outperforms the commercial solver within the time limit in terms of the solution quality and computational times, but it also obtains near-optimal solutions.

We also run the heuristic without the VNS improvement step, such that Algorithm 2 remains unchanged except for the exclusion of line 3. The results are presented under 'Heuristic without improvement' in Table 3. While the average computing time decreases substantially, an optimality gap of 58.6% is observed without the improvement step. In particular, for instance 6, optimality gap is observed as 205%. These highlight the importance of the VNS-based improvement heuristic in the construction of vehicle routes.

We also use our smaller instances to analyse the effect of including each neighbourhood structure in the heuristic approach. Table 4 shows the averages and 95% confidence intervals for percent increase in objective value and percent decrease in CPU time when each of the 7 neighbourhoods is excluded. Among these, $N_7$ (complete reconstruction of a truck tour) is most influential, with a loss of 5.66% in the objective value in its absence. This underlines the importance of the shaking step of the VNS, as this neighbourhood is only used for shaking. $N_4$ (drone to truck reassignment for the same depot) is also quite effective as it improves the objective value by an average of 3.24%. As expected, all remaining neighbourhoods have a positive but small effect on solution quality. From the 95% confidence intervals on CPU time changes, we observe that for all but one of the neighbourhoods, the change is statistically insignificant. The only exception is $N_4$, where a positive CPU time decrease is almost certain in its absence. On the other hand, given its significant improvement to the objective function, we retain this neighbourhood in the heuristic, as we do all other neighbourhoods, for the remainder of our experiments.

### 5.3. Computational results with case study instances

In this section, we implement the heuristic on 10 case study instances to analyse the structure of the heuristic solutions, compare our approach to various benchmarks, perform sensitivity analysis on important parameters, and evaluate the value of using a stochastic approach.

#### 5.3.1. Analysis of heuristic results

Table 5 displays the set of open depots, arrival times (in hours), and CPU times (in seconds) for the case study experiments. From the CPU times, we observe that the heuristic can find solutions to these instances within an average and maximum of 40 and 65 min, respectively. Given that these models are run in the pre-disaster stage, these computing times are acceptable for this context. On average, beneficiaries receive aid within 1.5 h of the start of operations, while the average makespan is 8 h over all instances. The table also provides the expected average arrival time at demand nodes visited by trucks or drones. In all instances, demand nodes served by drones have better expected average response times (1 h on average) compared to those visited by trucks (an average of 1.8 h). However, drone endurance poses a limitation to a better distribution of demand nodes between both vehicles.

Fig. 4 illustrates the heuristic result for Instance 10, while the road network and the candidate depot locations for this instance are shown in Fig. 5a. We note that the results for this instance are representative of those for other instances as well. The most frequently served demand nodes are indicated with the coverage area for each open depot across all scenarios. Demand locations are marked by their most frequently serving vehicle, and the shading of the vehicle further indicates the frequency of the assignment across the scenarios. Demand nodes that are further away from their allocated depot tend to be assigned to trucks. Conversely, the majority of the demand locations closer to their assigned depot are assigned to drones. Allocating further locations to truck is expected because of endurance and single visit constraints of drone. Furthermore, if we draw the smallest convex polygon encapsulating all open depots, demand locations within that region are mostly assigned to drones as most of these locations are drone eligible. Note that the circle around one of the open depots shows the approximate range for drone visits based on the endurance limit. The results for Instance 10 in Table 5 indicate that the expected average arrival time for trucks is higher than that for drones. This imbalance highlights the impact of drone flight endurance on vehicle allocation of demand nodes. Thus, increased endurance may result in more drone assignments and reduced waiting times.

**Table 5**
Set of open depots, arrival times (in hours), and CPU times (in seconds) for the case study experiments.

| Instance | Open depots | Average arrival time | Maximum arrival time | Average arrival time - trucks | Average arrival time - drones | CPU time |
|---|---|---|---|---|---|---|
| 1 | {0,1,2,9,11} | 1.37 | 7.19 | 1.57 | 1.08 | 2,364 |
| 2 | {0,1,2,11,13} | 1.50 | 10.17 | 1.80 | 1.05 | 3,824 |
| 3 | {1,3,8,9,10} | 1.60 | 8.17 | 1.93 | 1.05 | 2,062 |
| 4 | {0,3,8,9,13} | 1.40 | 6.51 | 1.66 | 1.08 | 2,215 |
| 5 | {2,4,8,9,14} | 1.52 | 7.9 | 1.92 | 0.95 | 2,111 |
| 6 | {0,1,2,5,7} | 1.49 | 7.26 | 1.87 | 0.89 | 2,245 |
| 7 | {2,7,10,11,13} | 1.45 | 6.79 | 1.71 | 1.05 | 1,718 |
| 8 | {6,9,10,11,12} | 1.51 | 8.13 | 1.82 | 0.93 | 3,880 |
| 9 | {0,3,6,7,8} | 1.62 | 9.14 | 2.01 | 0.95 | 1,787 |
| 10 | {1,4,6,9,14} | 1.68 | 8.79 | 2.13 | 0.91 | 1,811 |
| **Average** | | **1.51** | **8.01** | **1.84** | **0.99** | **2,402** |

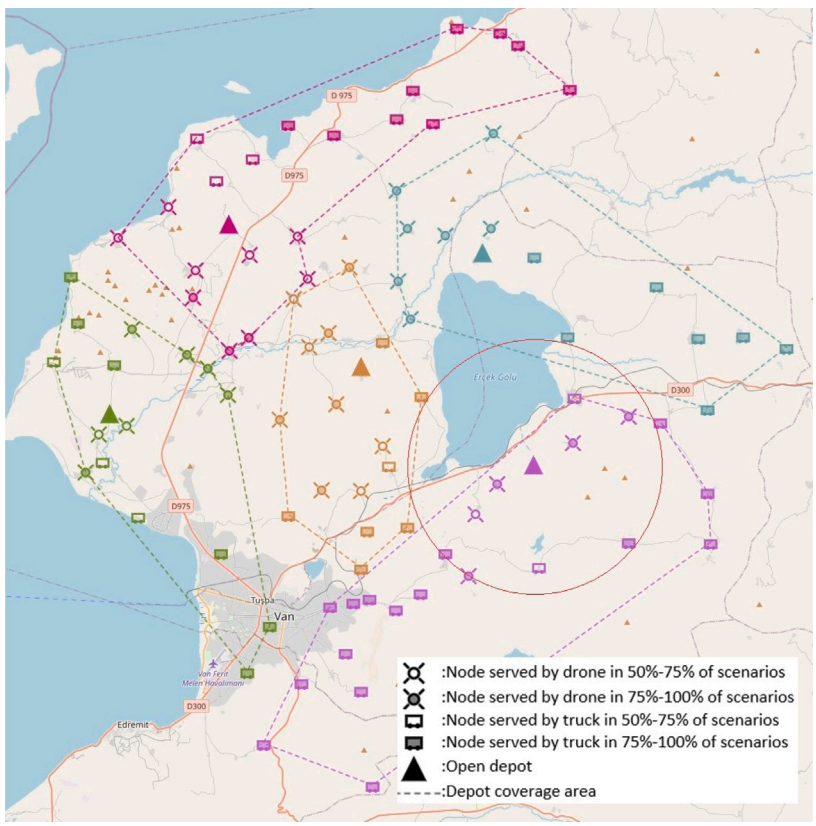


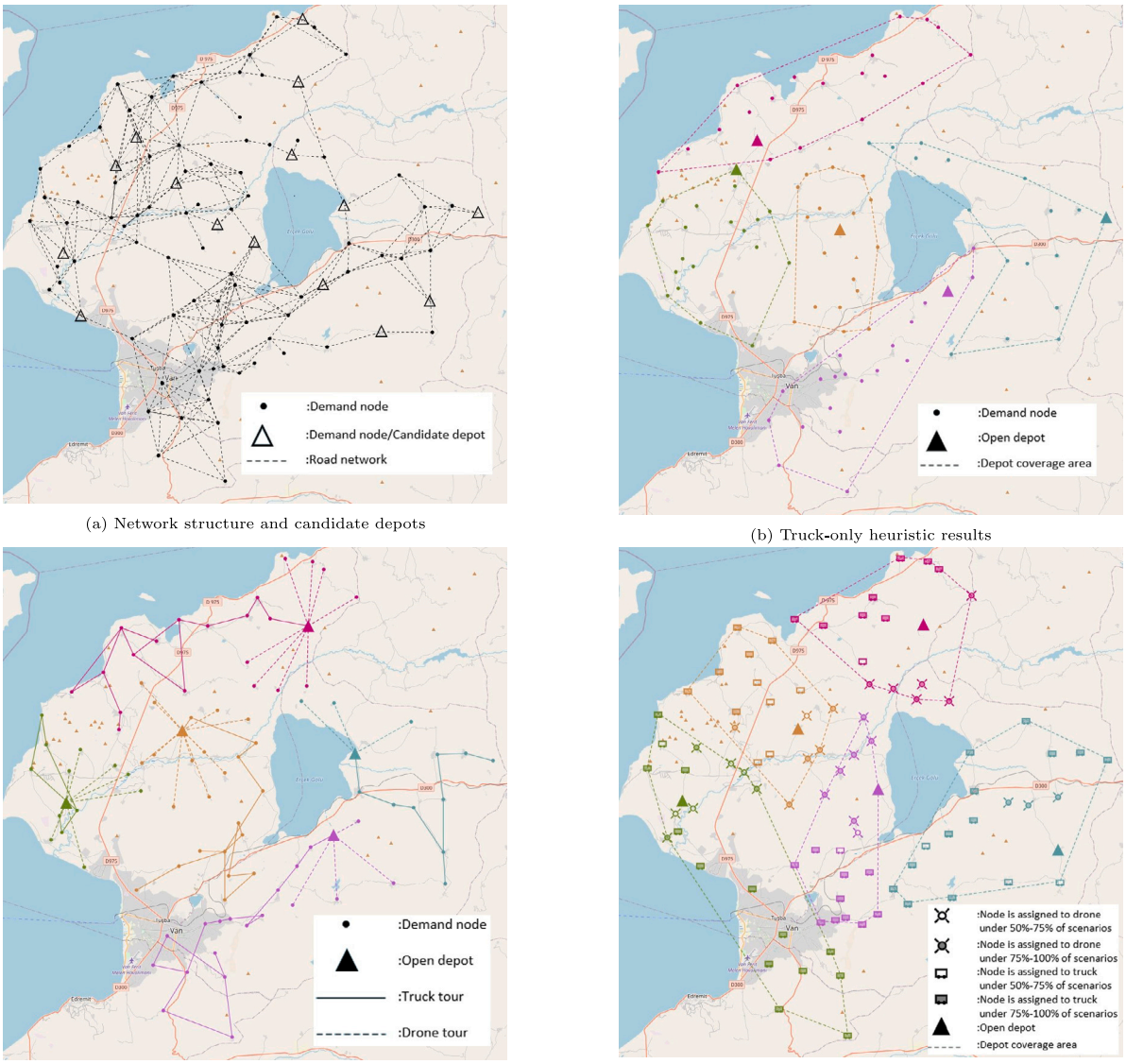

**Fig. 4.** Map demonstration of heuristic result for Instance 10: open depots, node–depot allocations and truck–drone assignment frequencies across all scenarios. Each colour indicates different depot and its assigned nodes.

### *5.3.2. Comparison to benchmarks*

Table 6 features comparative results derived from three benchmarks that use (i) only trucks to perform the deliveries, (ii) the heuristic with pre-disaster travel times, and (iii) the K-means algorithm for facility location. These benchmark cases are implemented on the same 10 case study instances, and their performances are compared against the baseline heuristic results (Table 5). For each benchmark, the set of open depots, the expected average arrival time (in hours), and the objective function gap relative to the baseline are reported.

Heuristic results for truck-only operations, as shown in Table 6, indicate an average gap of 116.8% across 10 instances compared to the baseline truck–drone results. This gap is calculated for each instance using the formula: $\%\text{Gap} = \frac{z_b - z_h}{z_h} \times 100\%$, where $z_b$ is the objective value of the benchmark (e.g., truck-only) case and $z_h$ is that of the baseline heuristic. The average of these gap values across all instances gives the reported 116.8%. In terms of the expected average arrival time across all instances, this translates approximately to the truck-only system requiring an average of 3.3 h per demand node, whereas truck–drone delivery significantly reduces the average arrival time to 1.5 h, as shown in Table 5. Fig. 5b demonstrates the corresponding heuristic results for Instance 10 on a map. In this particular instance, the locations of three open depots are the same as in the parallel truck–drone case with different coverage areas. Despite this, the fact that all demand nodes are serviced by trucks, as opposed to closer demand nodes being serviced by drones, results in more than double the expected latency.

To implement the second benchmark, we run the heuristic with a single scenario that uses the pre-disaster travel times on the road network. The performance of the identified depots in the post-disaster response is then evaluated using the DepotEvaluation algorithm across all disaster scenarios. This approach allows us to analyse the significance of damage assessment. From Table 6, incorporating road damage




(a) Network structure and candidate depots

(b) Truck-only heuristic results

(c) Truck-drone tours under pre-disaster conditions (no assessment case)

(d) Heuristic result with depots obtained from K-means

**Fig. 5.** Instance 10 map demonstrations. Each colour indicates different depot and its assigned nodes.

**Table 6**
Set of open depots, expected average arrival time ($z_b$) and % objective gap (% Gap) against heuristic solutions for benchmark cases.

| Instance | Truck-only operations | | | No road damage assessment | | | K-means depot location | | |
|---|---|---|---|---|---|---|---|---|---|
| | Open Depots | $z_b$ | % Gap | Open Depots | $z_b$ | % Gap | Open Depots | $z_b$ | % Gap |
| 1 | {0,2,9,11,14} | 3.1 | 125.0 | {0,1,2,4,10} | 1.6 | 14.6 | {0,8,9,11,14} | 1.8 | 30.4 |
| 2 | {1,3,6,8,14} | 3.4 | 124.6 | {1,2,3,5,6} | 1.7 | 14.8 | {0,1,2,11,13} | 1.5 | 0.2 |
| 3 | {5,6,7,12,14} | 3.3 | 105.9 | {1,5,6,7,13} | 1.7 | 5.9 | {1,3,8,10,14} | 1.6 | 0.9 |
| 4 | {4,8,11,12,13} | 3.5 | 149.8 | {0,1,3,9,13} | 1.4 | < 0.1 | {0,1,3,6,9} | 1.8 | 30.2 |
| 5 | {1,3,8,11,14} | 3.3 | 113.2 | {2,3,4,5,9} | 1.6 | 7.7 | {2,7,8,12,14} | 2.0 | 29.2 |
| 6 | {0,4,10,11,12} | 3.2 | 113.4 | {0,2,5,7,14} | 1.7 | 15.2 | infeasible | | |
| 7 | {2,6,7,10,13} | 3.2 | 119.0 | {2,6,7,11,13} | 1.5 | 1.5 | infeasible | | |
| 8 | {2,3,4,5,12} | 3.3 | 115.0 | {0,3,6,9,10} | 1.5 | < 0.1 | {0,1,2,3,5} | 1.6 | 5.2 |
| 9 | {0,3,7,12,13} | 3.4 | 106.7 | {0,2,3,6,12} | 1.6 | < 0.1 | {0,1,2,5,13} | 2.1 | 27.1 |
| 10 | {5,6,7,9,14} | 3.3 | 95.2 | {0,4,8,9,10} | 1.9 | 15.0 | {0,4,8,11,12} | 2.2 | 29.3 |
| **Average** | | 3.3 | **116.8** | | 1.6 | **7.4** | | 1.8 | **19.1** |

results in an average of 7.4% decrease in the expected objective, corresponding to a total of 10.5 h of response time, emphasising the critical role of damage assessment in disaster response. The demonstration of results based on pre-disaster conditions for instance 10 is shown in Fig. 5c with the corresponding truck–drone tours and open depots. The main difference from the baseline heuristic result given in Fig. 4 is the location of the open depot covering the northern part of the map. Considering the baseline map and the coverage area on the northern part, demand nodes closer to the fault lines between Van and Ercek Lake are assigned to drones. Conversely, in the no-assessment case, they are served by trucks, leading to the opening of a different depot.

To evaluate the importance of integrating pre-disaster location and post-disaster routing decisions, we compare our results against a benchmark where depot locations are determined simply based on geographical locations of the candidates, using a K-means clustering algorithm to locate the depots and assign demand nodes to open depots. We then use the DepotEvaluation algorithm for the identified depots and obtain the expected sum of the arrival times. On average, we observe a 19.1% increase in the expected objective when a K-Means approach is used. Furthermore, K-Means results in infeasible truck–drone tours for Instances 6 and 7 due to drone endurance constraints. We illustrate the K-Means result for Instance 10 in Fig. 5d, where open depots and the coverage areas are significantly different from those in the baseline results. K-Means results in a more homogeneous distribution of open depots on the map, as expected. However, the demand nodes closer to fault lines are most frequently distributed among three open depots, whereas they are served by four open depots in the baseline.

#### 5.3.3. Sensitivity analysis

In this section, we evaluate the effects of varying drone endurance, drone speed, and number of open depots on the expected average response time, providing insights into their impact on operational efficiency.

First, Fig. 6a depicts the impact of varying the number of open depots on the expected average arrival time across 10 instances. The results show the average arrival time varies between 3.4 and 0.4 h for 3 and 15 open depots, respectively. Each additional depot, equipped with a truck and a drone, contributes to significantly better response times, albeit at a diminishing rate. This suggests a trade-off between the investment cost and the response time, providing insights for the decision-making process regarding the optimal number of open depots to employ.

Apart from the baseline flight endurance of 30 min, we also explore the effect of longer flight endurance, specifically 1 h, 2 h, and unlimited time. Small delivery drones generally have shorter endurance than 1 h (JD, 2018; The Sun, 2022), but we consider higher endurance values considering future developments in drone technology. Fig. 6b shows the impact of the varying endurance on the expected average arrival time, illustrating the impact of extended battery life on reducing response times. Extending the endurance from 30 min to 1 h decreases the average arrival time by 18 min, by 20%. However, an additional hour increase beyond 1 h yields only marginal improvement in the objective, around 1%. Moreover, transitioning to an unlimited battery life does not yield an improvement in the arrival time compared to the two-hour limit. In more densely located areas, the impact of improved battery life on response times might be even less pronounced. These findings indicate that even the current drone technology can be beneficial to achieve timely humanitarian aid delivery.

Lastly, we explore varying drone speeds as considered by Murray and Chu (2015), specifically 24 km/h (15 mph) and 56 km/h (35 mph), as well as the maximum drone speeds reported by some big retailer companies (JD, 2018; The Sun, 2022), which reach up to 100 km/h. We use 72, 88 and 104 km/h to represent these reported speeds. These are compared to the baseline drone speed of 40 km/h (25 mph). Fig. 6c illustrates the change in the expected average arrival time across 10 instances. Although subject to diminishing returns, an increase in drone speed significantly improves the response time. Specifically, drones operating at 56 km/h reduce the average arrival time by 24 min compared to our baseline drone speed, corresponding to a 26% improvement.

#### 5.3.4. Value of stochastic solution and expected value of perfect information

As our last analysis, we calculate the Value of Stochastic Solution (VSS) and the Expected Value of Perfect Information (EVPI) to assess the value of a stochastic modelling approach. We consider the same 10 case study instances used in the heuristic analysis, as presented in Section 5.3 and Table 5. Let $z_h$ denote the expected average arrival time across all instances, as obtained from the heuristic analysis in Table 5.

VSS indicates the expected gain obtained from incorporating uncertainties into the model and using a stochastic modelling approach, rather than solving its deterministic counterpart based on the expected values of the uncertain parameters. In other words, it shows how much worse the solution would be if average values were used instead of considering uncertainties in the decision making. The deterministic counterpart of our problem has travel times equal to the expected travel times on the road network, which defines the Expected Value Problem (EVP). We solve the EVP with our heuristic as if there were a single scenario to find the depot locations. Then, the post-disaster truck–drone routes under all scenarios are generated by DepotEvaluation algorithm for the resulting depot locations. Following this method, we obtain the expected average arrival times when EVP is solved, denoted by $z_{EVP}$. The difference in the expected objective values of the stochastic model, $z_h$, and the deterministic model where the uncertainties are replaced with average values, $z_{EVP}$, gives the VSS. The relative VSS (%VSS) is then calculated as: $\%VSS = \frac{z_{EVP}-z_h}{z_h} \times 100\%$. For our instances, the average VSS across all instances (%VSS) is obtained as 3.4%. Although the average gain appears small, it corresponds to a total of 4.5 h of improvement in the response time across all nodes. Furthermore, the maximum %VSS across all instances is obtained as 17.4%, indicating a total improvement of 24.5 h. Considering the importance of timely delivery, using a stochastic modelling approach for preparedness stage decisions can improve the performance of post-disaster operations.

Expected Value of Perfect Information (EVPI) shows the maximum improvement one would obtain from having the perfect information. It corresponds to the expected gain when decisions are made under known future information. We treat each disaster scenario as a known future scenario and give depot location decisions based solely on the post-disaster routing for this scenario. To do so, we run the heuristic for only the scenario under consideration. The overall expectation across all scenarios gives the expected value of decision-making with perfect information, denoted by $z_{PI}$. On the other hand, we have the expected objective value from the stochastic model, $z_h$. Their difference is denoted by EVPI, which states the maximum gain from having perfect knowledge. The relative value is calculated as follows: $\%EVPI = \frac{z_h-z_{PI}}{z_{PI}} \times 100\%$. Across the 10 instances, we obtain a %EVPI of 7.6%, which corresponds to 10 h. The maximum %EVPI across all instances results in a difference of 20 h in response time. These findings suggest that having better knowledge and a good identification of the disaster can significantly improve the response time in the post-disaster stage.

## 6. Conclusions and further research directions

Road networks are essential to achieve timely response in the post-disaster and disasters can severely disrupt the road networks and hinder the humanitarian response activities. The uncertain nature of the disasters makes it challenging to plan and execute timely response operations. In this paper, we consider parallel truck–drone deliveries from depots to deliver light-weight humanitarian aid bundles to beneficiaries. We define a stochastic network design problem to locate the depots in the pre-disaster while also considering the post-disaster routing decisions. Road network vulnerability is specifically addressed via inflated travel times to reflect disaster-related uncertainties. Drones, in addition to trucks, are involved to make aid delivery to overcome the longer travel times and achieve better response times.

We model the stochastic location-routing problem with trucks and drones as a two-stage stochastic programming model. The objective

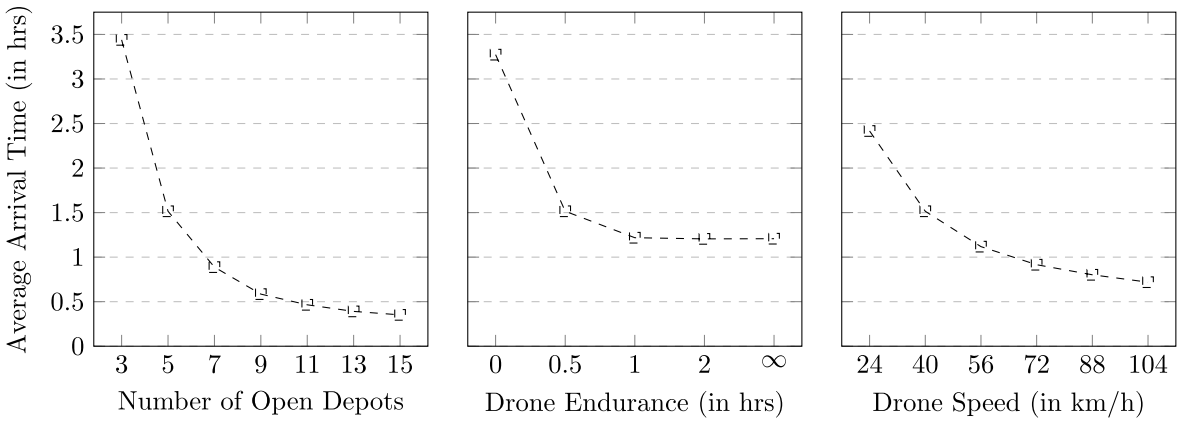



**Fig. 6.** Relationship between the average objective value and varying (a) number of open depots, (b) drone endurance, and (c) drone speed.

is to minimise the expected average waiting times of the demand locations. Our experiments show that Gurobi cannot solve the problem within a reasonable time limits even on small-sized problems. Thus, we develop a problem-tailored heuristic based on local search and VNS for location and routing decisions, respectively. Optimality gaps are shown to be very small on small-sized problems, and the proposed heuristic can solve large instances within reasonable times. Our experiments on a case study indicate that drones can significantly reduce the response times compared to a truck-only system, with the expected average arrival time decreasing from 3.3 h to 1.5 h, marking an improvement of approximately 55%. The proposed approach can potentially help humanitarian agencies make strategic decisions such as location and pre-positioning of relief items, equipment, and vehicles, and evaluate the effect of integrating drones into their fleet on disaster-prone regions.

In this paper, we consider parallel truck and drone deliveries from depots. This structure is mainly reasonable when most of the demand locations are drone-eligible such that drones can provide rapid deliveries from depots within the endurance limit. However, if most of the demand locations are not eligible, the extension of the current stochastic location-routing problem with a synchronised truck–drone delivery model must be considered to expand the drone service area. This would allow drone utilisation in that scenario and provide more flexibility even on the current considered problem environments and improve the response times further. Furthermore, comparison of different truck–drone structures on different problem instances can provide further managerial insights. Different uncertainties other than the road vulnerability can also be incorporated into the network design problem such as uncertain demand requests from demand locations or demand quantities. Another practical extension is the use of multiple vehicles of the same type in each depot. Introduction of such properties would make the problem even more complex and necessitate efficient and effective solution approaches.

## CRediT authorship contribution statement

**Hannan Tureci-Isik:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Melih Çelik:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Ece Sanci:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ejor.2025.08.057.

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